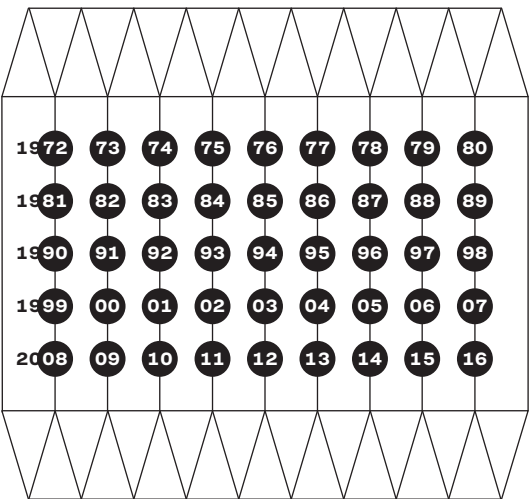


UNCHARTED

Footnotes to the Atlas

AN UNSTRUCTURED AND EXPANDING
COMMENTARY OVER THE PRODUCTION
OF CONTEMPORARY CARTOGRAPHY.
THIS PRINTOUT IS PART OF
UNCHARTED, A PROJECT BY
FOLDER FOR **AFTER BELONGING**,
OSLO ARCHITECTURE TRIENNALE,
SEPT. 8TH — NOV. 27TH, 2016.



1.

Equatorial countries declare that the geostationary synchronous orbit is a physical fact linked to the reality of our planet because its existence depends exclusively on its relation to gravitational phenomena generated by the earth, and that is why it must not be considered part of the outer space. Therefore, the segments of **geostationary synchronous orbit** are part of the territory over which Equatorial states exercise their national sovereignty. The geostationary orbit is a scarce natural resource, whose importance and value increase rapidly together with the development of space technology and with the growing need for communication; therefore, the Equatorial countries meeting in Bogota have decided to proclaim and defend on behalf of their peoples, the existence of their sovereignty over this natural resource.

SOURCE

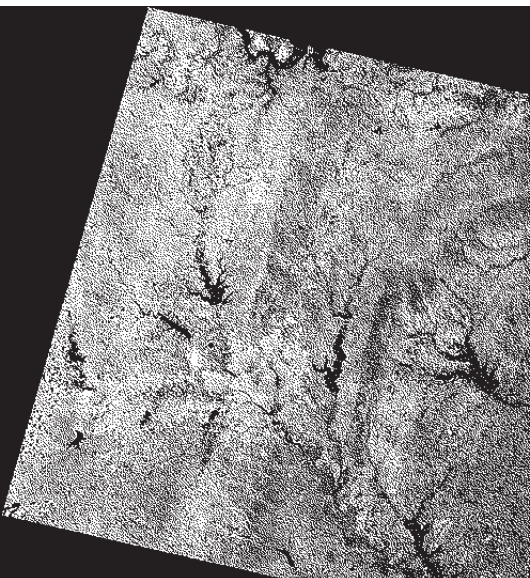
Excerpt from the *Declaration of the First Meeting of Equatorial Countries in Bogotá* (November 26th–December 3rd, 1976), adopted on December 3rd, 1976.

2.

Recent analyses of the U.S.-held Landsat data record have disproved the general assumption that the 36+-year record provides **uninterrupted coverage**, or even global coverage on an annual basis. In 2002, the National Satellite Land Remote Sensing Data Archive (NSLRSDA) at the Earth Resources Observation and Science (EROS) Center, generated annual and seasonal coverage maps for each Landsat sensor—Multispectral Scanner System (MSS), Thematic Mapper (TM), and Enhanced Thematic Mapper Plus (ETM+). The results astounded even those who had worked on the Landsat program for decades. Holes in the coverage maps were often for well-understood reasons, such as equipment failures, availability of downlink opportunities, or political or management issues. But other holes were not so easily explained.

SOURCE

From *Preserving Landsat's Legacy* by Terry Arvidson, in "Pecora 17 – The Future of Land Imaging... Going Operational", November 18-20, 2008 Denver, Colorado.



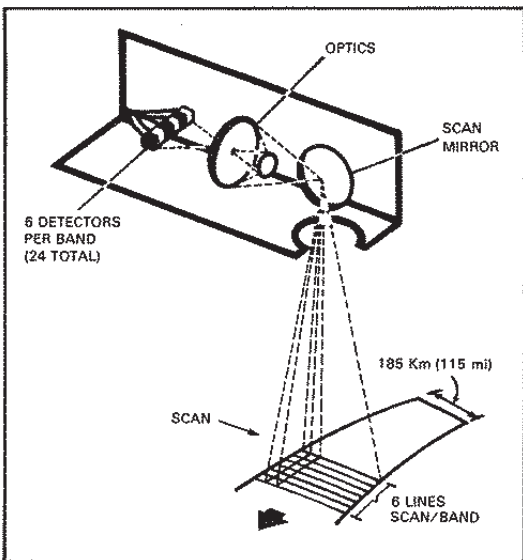
Dallas area, Texas, July 25, 1972
 resolution : 60 mt/px
 false-color image

As the **imagery** scrolled into view, there were clouds...and then more clouds...and then land. But something looked strange. A technician in the room lamented: “It’s terrible. It has moiré patterns.” But after figuring out the location of the image, scientists realized that the imagery was showing circular outcrops of the ancient fold mountains.

Any lingering anxiety in the room quickly turned to excitement, and **skeptics** were promptly converted into supporters. The well-known U.S. Geological Survey cartographer Alden Colvocoresses, who had been cynical about any cartographically accurate data being collected with “ a little mirror in space,” turned to his colleagues after seeing the MSS imagery and said: “gentlemen, **that’s a map.**”

SOURCE

Excerpt from *First of three million*, July 2012
earthobservatory.nasa.gov



MSS Scanning arrangement.

Landsat sensors record only a small portion of the total radiation reflected or emitted from the Earth's surface. However, the visible and infrared spectra carry most of the information of interest in **surveying natural resources** and the environment.

SOURCE

The Landsat Story, Module U-2 by National Aeronautics and Space Administration, Office of Space and Terrestrial Application, January 1980

5.

GABRYNOWICZ: What **value**, if any, do the [remote sensing] **Principles** have in your mind?

HOSENBALL: The biggest value that it has is it presses for **international cooperation** in remote sensing throughout, in various Principles. There is a push for international cooperation. It calls attention to states that this technology exists. There is someplace you can look to see what your rights are, what your access is, as well as other provisions like dispute [and consultation] provisions.

SOURCE

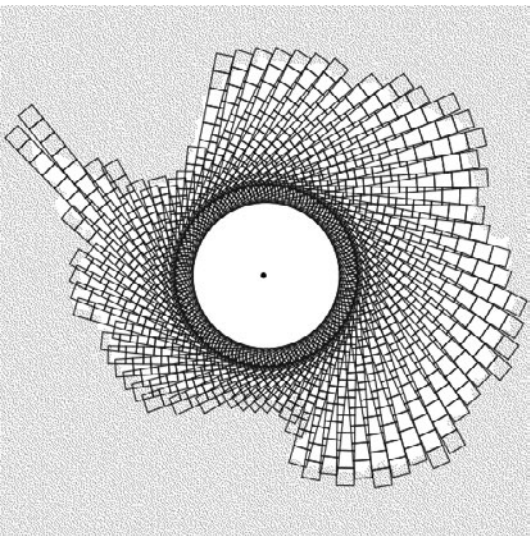
Excerpt from *UN Principles Relating to Remote Sensing of the Earth from Space, a Legislative History - Interviews of Members of the United States Delegation*, by Joanne Irene Gabrynowicz, editor, 2002

6.

To be effective in **monitoring global change** or in supporting resource management, the delivery of high-quality, well-calibrated, remotely sensed data must be sustained over long periods. Certain data sets, such as those related to Earth's radiation budget, should be **acquired continuously** over decades. In some cases, data must also be delivered with few or no gaps in the operation of the satellites.

SOURCE

Excerpt from *The Future of Remote Sensing From Space: Civilian Satellite Systems and Applications*, U.S. Congress, Office of Technology Assessment, July 1993



This map represents the extent of the **landsat mosaic** [on the Antarctica territory]. Because of landsat 7's **near-polar orbit** there is a “hole” at the south pole.

SOURCE

Landsat Mosaic of Atarctica, USGS February 2007, <http://landsat.gsfc.nasa.gov/?p=483>

8.

The orbit of Landsat 7 is repetitive, circular, **Sun-synchronous**, and near polar at a nominal altitude of 705 km (438 miles) at the Equator. The spacecraft crosses the Equator from north to south on a descending orbital node from between 10:00 AM and 10:15 AM on each pass. Circling the Earth at 7.5 km/sec, each orbit takes nearly **99 minutes**. The spacecraft completes just over **14 orbits per day**, covering the entire Earth between 81 degrees north and south latitude every 16 days.

SOURCE

Excerpt from *Landsat 7 Science Data Users Handbook*, National Aeronautics and Space Administration, 2016

9.

GABRYNOWICZ: What were the first serious discussions that you remember?

HOSENBALL: Well, the serious discussions basically boiled down to [prior] consent and sovereignty. (...) The Soviets and French submitt[ed] a paper which said basically that (...) no objection to sensing from space but you had to get prior consent from a sensed state before you can make it available to a third state, which meant that (...) if they didn't announce which state had been sensed, nobody would know that fact and therefore they wouldn't give it to the sensed state. (...) The developing countries didn't want you to either sense or distribute the data to anyone. (...) The U.S. position was obviously free flow of information across national borders. They [less developed countries] were also afraid, not knowing as much about the technology, that they might be harmed by it in some way. They wanted to be helped by it (...) [But] nothing in these Principles allows somebody to enter into your territory without your consent.

SOURCE

Excerpt from *UN Principles Relating to Remote Sensing of the Earth from Space, a Legislative History - Interviews of Members of the United States Delegation*, by Joanne Irene Gabrynowicz, editor, 2002

10.

GABRYNOWICZ: (...) Let me ask you if the following statement is correct: The **non-discriminatory access** policy was the U.S. answer to both the **prior consent** issue and the issue of **third party distribution**.

HOSENBALL: I think that's true. (...) Of course, the many countries arguing in favor of prior consent do not have space programs and cannot prevent the United States from taking data from space.

SOURCE

Excerpt from *UN Principles Relating to Remote Sensing of the Earth from Space, a Legislative History - Interviews of Members of the United States Delegation*, by Joanne Irene Gabrynowicz, editor, 2002

11.

Irrespective of which model is adopted, such a model should institute an inviolable mechanism that would ensure (except in cases of emergencies such as natural disasters) **rapid, non-discriminatory and non-discretionary acquisition, processing and distribution of data**, and should forbid, under all circumstances, the auctioning of orbits to the highest bidder.

SOURCE

Excerpt from *International remote sensing system: a possibility* by Adigun Ade Abiodun, Office for Outer Space Affairs, United Nations, New York

12.

Although the **Land Remote Sensing Policy Act of 1992** reversed the **privatization** track for Landsat and returned the satellite system to the federal government, the act also authorized the Secretary of Commerce to license operators of private remote sensing space systems. It allowed the operators to use their data as they wish, including choosing their customers and offering their data at prices that vary by customer.

SOURCE

Excerpt from *Landsat: Overview and Issues for Congress*, Peter Folger, Specialist in Energy and Natural Resources Policy, October 27, 2014

13.

Evolving views over the **public or private** nature of the program were influenced by factors other than funding. One observer identified four factors:

1. Landsat data proved important in planning **U.S. military operations** in the 1992 Gulf War.
2. **Other countries** had launched similar land remote sensing satellites, and these spacecraft (...) were perceived as possible challenges to the U.S. stake in the international market for remote sensing data
3. Growing interest in global **climate change** and its effects on the Earth's surface led scientists to increasingly value time-series data from a consistent platform in space for identifying environmental changes.
4. The difficulties of **commercializing** the Landsat system became clear, and federal agencies perceived that private companies might not be able to provide equivalent data at the scale the agencies required.

SOURCE

Excerpt from *Landsat: Overview and Issues for Congress*, Peter Folger, Specialist in Energy and Natural Resources Policy, October 27, 2014

14.

GABRYNOWICZ: Do the Principles apply to **military satellites**?

HOSENBALL: No.

GABRYNOWICZ: What was the position of the U.S. on that?

HOSENBALL: [Though the question of military satellites was never publicly raised by any country,] our position (...) was that it was remote sensing of the Earth environment and its **natural resources**. That was the definition of remote sensing, and that's what those **Principles** [would] apply to.

SOURCE

Excerpt from *UN Principles Relating to Remote Sensing of the Earth from Space, a Legislative History - Interviews of Members of the United States Delegation*, by Joanne Irene Gabrynowicz, editor, 2002

15.

Data from civilian satellites systems such as Landsat (...) have considerable **military utility**. (...) For example, the use of Landsat and SPOT data gave the U.S. and its U.N. allies a marked advantage over Iraq in the **Persian Gulf Conflict**. The U.S. Defense Mapping Agency used these data to create a variety of maps for the U.S.-led battle against Iraqi forces.

SOURCE

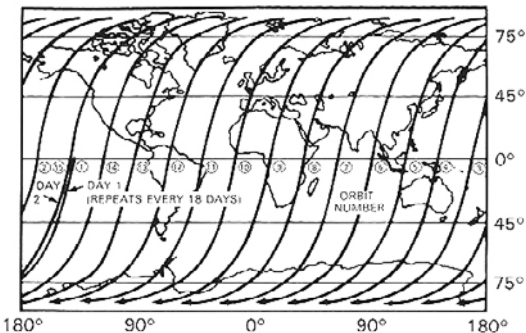
Excerpt from *The Future of Remote Sensing From Space: Civilian Satellite Systems and Applications*, U.S. Congress, Office of Technology Assessment, July 1993

16.

The primary elements of International Remote Sensing System (IRSS) should consist of (I) a number of complementary Earth observation satellites, with **no national identity**, but with adequate capabilities to meet the environmental and natural resources information needs of the **international user community**, and (II) a set of data receiving and processing ground stations. The system should be wholly managed by an international consortium.

SOURCE

Excerpt from *International Remote Sensing System: A Possibility* by Adigun Ade Abiodun, Office for Outer Space Affairs, United Nations, New York



Typical Landsat **daily ground trace** (daylight passes only). The sun-synchronous orbit feature is very advantageous because of constant illumination is helpful in the detection of changes in a scene, and low/high sun angles are useful in studying landscape and terrain features.

SOURCE

From *The Landsat Story, Module U-2*
by National Aeronautics and Space
Administration, Office of Space and
Terrestrial Application, January 1980

Over the last 15 years, the **military use** of Landsat has also increased. At the end of the Cold War, few could visualize the unstable world we live in today, with terrorist attacks occurring at locations throughout the world. The **need for reconnaissance** has not diminished as projected, but has greatly changed. Instead of focusing on specific sites in one or two large countries (USSR and China), reconnaissance must now be dispersed around the world, and new technologies are required. Since Desert Storm, the military has recognized the usefulness of moderate resolution data to provide a wide synoptic view. In 1998 NASA contracted with Earthsat to create a world wide mosaic of Landsat imagery (called GeoCover) (Tucker et al., 2004). NGA followed this program by partially funding the development of Geocover LC, which is a land-cover and land-use map that Earthsat created from the GeoCover imagery. As a result, the military has become one of the biggest users of Landsat imagery.

SOURCE

From *Landsat in Context: The Land Remote Sensing Business Model* by Kass Green, in "Photogrammetric Engineering & Remote Sensing", Vol. 72, No. 10, October 2006, pp. 1147–1153.

19.

If there was a commercial market of the size required to support the design, construction, launch, operation, and archiving of moderate resolution satellite data, then the **commercial sector** would be providing the data, and there would be no role for the government in the provision of moderate resolution data. However, because of **high fixed costs**, commercial land satellite remote sensing at any resolution is not, and will not soon be, a viable non-military commercial venture. The requirements of commercial or government civilian users cannot support the investment necessary to build, launch, and operate an earth observing satellite.

SOURCE

From *Landsat in Context: The Land Remote Sensing Business Model* by Kass Green, in "Photogrammetric Engineering & Remote Sensing", Vol. 72, No. 10, October 2006, pp. 1147–1153.

Landsat's base of support has been shallow, narrow, and fragmented. The **user base** was initially made up of scientists and the difficulty of working with the data slowed its adoption for many years. Then, seven years of **ill-conceived "commercialization"** resulted in Landsat prices and licensing terms which stagnated the adoption of the imagery. Without a strong user base, the institutional framework supporting Landsat has been weak from the beginning. The program has bounced from one agency to another every five to eight years. It has only been since the launch of Landsat-7 in April 1999 that the use of Landsat data has broadened widely to embrace many sectors including the military; and it is ironic that it is now that the Program's future is again at risk. Be it food security, disaster preparedness, urban planning, endangered species protection, water quality, or battle field situational analysis, the uses of Landsat data, and particularly Landsat in combination with high resolution data, have exploded and the user base is becoming politically motivated.

SOURCE

From *Landsat in Context: The Land Remote Sensing Business Model* by Kass Green, in "Photogrammetric Engineering & Remote Sensing", Vol. 72, No. 10, October 2006, pp. 1147-1153.

The technology to procure high resolution imagery from space has been available to the **U.S. military** for at least four decades, but available to **civilian users** for only the last five years. The world we live in today, with abundant and minimally restricted access to **high resolution imagery** worldwide, is relatively new and somewhat threatening to defense and intelligence organizations. Given the roots of remote sensing technology in the requirements of the military, it is perhaps surprising that the technology has been allowed to migrate to civilian markets at all. This nation's fundamental belief in democracy and capitalism and in the transparency upon which both are based, will always create tension between the desire to conceal covert activities while simultaneously promoting open information exchange.

SOURCE

From *Landsat in Context: The Land Remote Sensing Business Model* by Kass Green, in "Photogrammetric Engineering & Remote Sensing", Vol. 72, No. 10, October 2006, pp. 1147–1153.

To comprehend the **political and policy dynamics** of **commercial remote sensing** (CRS), it is helpful to view remote sensing as a large integrated system that extends beyond just the technical aspects of the satellites themselves. One can almost visualize remote sensing as a giant organism that requires various parts to efficiently function at an optimum level. Similar to the concept of network theory and large technological systems, such as electrical power generation and distribution systems, CRS can be viewed as a **super system** that comprises hardware (satellites, sensors, ground stations, launch vehicles, communications and control systems); software (imagery processing, distribution, and analysis programs); data (images and value-added products), people (system operators, policymakers, regulators, scientists, national security personnel, civil/commercial data users); control and/or support mechanisms (CRS laws, policies, regulations, government contracts); and its supporting or competing environments (market conditions, insurance, foreign satellite systems, etc.).

SOURCE

From *A Political History of U.S. Commercial Remote Sensing, 1984-2007: Conflict, Collaboration, and the Role of Knowledge in the High-Tech World of Earth Observation Satellites* by Kenneth P. Thompson, Dissertation submitted to the Faculty of the Virginia Polytechnic Institute and State University in partial fulfillment of the requirements for the degree of Doctor of Philosophy in Science and Technology Studies, November 20th, 2007

23.

Space-based remote sensing offers us the advantage of **prosthetic vision** and the chance to use radiant energy from the hidden regions of the electromagnetic spectrum to revolutionize our understanding of the shifting realities of life and the existence of planet Earth.

SOURCE

Mark Elam, *Spyglass On The World: The Culture and Politics of Remote Sensing*, draft paper presented at the Cultural Politics of Technology Workshop, Trondheim, Norway, Centre for Technology and Society, Norwegian University of Technology and Science, June 15–16, 1998.

To gain a sense of the power and influence of earth observation satellites on **politics, economics, and ideology**, it is helpful to view these fascinating remote sensors in their geopolitical and geoeconomic contexts. Earth observation satellites acquire information on a global scale, transcending the limits of borders and **national sovereignty**. These prosthetic eyes in space are able to penetrate denied areas with capabilities that exceeded their predecessor balloon and airbreathing cousins (aerial photography). In a geopolitical context, civil and commercial observation satellites were used as a political tool—peering into the inner confines of states involved in weapons proliferation, human rights abuses, border disputes, and other activities that were once thought protected by physical boundaries and geophysical space. However, with the advent of spaceborne remote sensors, terrestrial borders and vast physical spaces could not longer inhibit this **transparency**.

SOURCE

Kenneth P. Thompson, *A Political History of U.S. Commercial Remote Sensing, 1984-2007: Conflict, Collaboration, and the Role of Knowledge in the High-Tech World of Earth Observation Satellites*, Dissertation submitted to the Faculty of the Virginia Polytechnic Institute and State University in partial fulfillment of the requirements for the degree of Doctor of Philosophy in Science and Technology Studies, November 20th, 2007.

In the case of Landsat, geoimagery was used to promote U.S. ideals of transparency and societal openness in contrast to the Soviet-block's views on a closed society. This was done through the deployment of several ground stations in various regions of the world and making Landsat data available on a non-discriminatory basis to all nations that operated those stations and to other nations to which the ground-station operators chose to sell Landsat imagery. According to Joanne Gabrynowicz, the Landsat system was originally used a geopolitical tool to demonstrate the superiority of U.S. technology in geoimaging and to promote the ideal of sharing information about the earth's surface and its environment and natural resources.

SOURCE

Kenneth P. Thompson, *A Political History of U.S. Commercial Remote Sensing, 1984-2007: Conflict, Collaboration, and the Role of Knowledge in the High-Tech World of Earth Observation Satellites*, Dissertation submitted to the Faculty of the Virginia Polytechnic Institute and State University in partial fulfillment of the requirements for the degree of Doctor of Philosophy in Science and Technology Studies, November 20th, 2007.

In 1996, international politics entered the U.S. commercial remote sensing (CRS) policy arena in a significant way when Israel began lobbying the USG concerning the future technical capabilities or operations of U.S. high-resolution satellites. At the time, three U.S. firms had already been granted licenses by DOC (NOAA) to operate 1-meter panchromatic CRS satellites. Israel was particularly concerned that high-resolution imagery of its key military installations and military movements within its borders would compromise its security and make it vulnerable to attacks by its adversaries. Such concerns began shortly after the passage of the Land Remote Sensing Policy Act in 1992. One major point of concern was the United Arab Emirates' desire to purchase a CRS satellite system from a U.S. Company, Litton Itek, in late 1992. Israeli concerns were further exacerbated when a Saudi firm, EIRAD, sought to acquire a major financial interest in the Eyeglass venture.⁹⁷ As a consortium involving Orbital Sciences, GDE, and Itek, Eyeglass International had received a license for a 1-meter system from DOC in 1994. [...] After a significant amount of discussion and diplomatic jockeying between U.S. and Israeli officials, the U.S. Senate passed an amendment (known as the Kyl-Bingaman Amendment and discussed in more detail in a subsequent chapter) to the 1997 Defense Authorization Act that would restrict U.S. CRS licensees from imaging Israel or other designated areas unless the imagery was no better than what was offered by other non-U.S. commercial satellite systems. [...] The 2-meter agreement between the U.S. and Israel was a **shutter control-like measure**, albeit permanent in nature and favoring a single country.

SOURCE

Kenneth P. Thompson, *A Political History of U.S. Commercial Remote Sensing, 1984-2007: Conflict, Collaboration, and the Role of Knowledge in the High-Tech World of Earth Observation Satellites*, Dissertation submitted to the Faculty of the Virginia Polytechnic Institute and State University in partial fulfillment of the requirements for the degree of Doctor of Philosophy in Science and Technology Studies, November 20th, 2007.

ALL OF THE FOOTNOTES
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ON NOVEMBER 28TH, 2016.